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PERFORMANCE SPECIFICATION

ZINC-NICKEL ELECTROPLATING FOR FASTENERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers requirements for an electrodeposited zinc-nickel (Zn-Ni) alloy plating, applied in an alkaline process, with a hexavalent chromium-free conversion coating and supplementary top coat or sealant. This MIL specification selects classification codes from ASTM F1941/F1941M and details additional requirements, such as plating composition and torque/clamp force behavior of the plating on surrogate threaded fasteners. Although developed for threaded steel fasteners, this plating may be used on other steel products. While other plating thickness and supplementary treatment options are available for unique applications, the requirements of this specification are limited to the ASTM F1941/F1941M codes specified herein. This specification is intended for use on new fasteners and fasteners which previously may have specified coatings containing cadmium or hexavalent chromium, substances which have been added to the Aerospace and Defense Declarable Substances List ("AD-DSL"). This specification was developed to address steel fastener coatings without hazardous substances for use on military ground vehicles. For other applications, users should consult the appropriate cognizant engineering organization.

1.2 Plating of internal diameters. When process limitations will not allow adequate plating to be deposited on internal diameters requiring corrosion protection, an alternate method of protection should be specified in a drawing or acquisition document.

1.3 Classification. Classification of this zinc-nickel plating is consistent with the requirements of ASTM F1941/F1941M (see 3.2) with the addition of a Torque/Clamp Force Type.

Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: Standardization: FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Detroit Arsenal, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.3.1 Type. The Torque/Clamp Force K-Factor requirement for the plating, as tested per 4.6, is specified as one of the following types (see 6.4 regarding how the torque/clamp force type is to be interpreted for fastener assembly):

Type K16: 0.14 – 0.18 K-Factor
Type K20: 0.18 – 0.22 K-Factor

NOTE: For threaded fasteners without a type specified, K20 is the default type. For all other parts without threads requiring friction control in the plating, the type should be specified.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.6).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.24	-	Part Identifying Number (PIN) Code System Standard for B18 Fastener Products

(Copies of this document are available online at https://www.asme.org.)

ASTM INTERNATIONAL

ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B571	-	Standard Practice for Qualitative Adhesion Testing of Metallic Coatings
ASTM F1470	-	Standard Practice for Fastener Sampling for Specified Mechanical Properties and Performance Inspection
ASTM F1941/F1941M	-	Standard Specification for Electrodeposited Coatings on Mechanical Fasteners, Inch and Metric

(Copies of these documents are available online at https://www.astm.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4014	-	Fasteners - Hexagon Head Bolts - Product Grades A and B
ISO 4032	-	Hexagon Regular Nuts (Style 1) - Product Grades A and B
ISO 16047	-	Fasteners - Torque/Clamp Force Testing

(Copies of these documents are available online at https://www.iso.org.)

SAE INTERNATIONAL

SAE AMS2461	-	Plating, Zinc-Nickel Alloy (12 to 16% Ni)

(Copies of this document are available online at https://www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Responsibility for compliance. The inspection set forth in this specification shall become a part of the plating applicator's overall inspection system or quality program. Any changes made to the quality plan or plating process shall be subject for re-approval (see 4.1).

3.1.1 Records. The plating applicator shall maintain a record of the history of each processing bath and the results of all chemical analyses performed for government review (see 4.1.2 and Table I).

3.2 Plating process. Unless otherwise specified in the contract, the plating shall be electrodeposited in an alkaline process[1] and meet the requirements of ASTM F1941/F1941M Classification Code Fe/Zn-Ni 8BNS or 8BNSL (the first digit in the plating code after Fe/Zn-Ni indicates the minimum plating thickness in microns (8), the next two digits indicates the passivation type, and the last digits indicate additional topcoat requirements) except requirements as specified herein shall take precedence (see 4.2).

[1] An alkaline process must be used for threated fasteners with a specified maximum core hardness exceeding HRC 39. For all other products, an acid zinc-nickel plating process may be utilized with approval by the cognizant engineering organization.

3.2.1 Replating. Parts having specified maximum core hardness 39 HRC and below may be stripped and reprocessed per this MIL-PRF-32647 specification.

3.2.2 Use of brighteners. As noted in SAE AMS2461, "brighteners used as additives to the plating solution shall not be permitted for use on steel parts 36 HRC and higher or on threaded steel fasteners 34 HRC or higher" (reprinted with permission from SAE AMS2461; © SAE International. All rights reserved).

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3.3 Plating composition. The zinc-nickel alloy, as plated, shall be 12% – 16% nickel by weight (see 4.3).

3.4 Sealant or topcoat. The plating applicator shall apply the sealant or topcoat necessary to achieve the performance requirements of this specification. The friction-modifying lubricant shall be integral to the sealant or topcoat or added as an extra coating material layer if necessary. The final finish shall be dry-to-touch (see 3.9 and 4.9).

3.5 Corrosion resistance. Corrosion resistance testing shall be conducted in accordance with ASTM F1941/F1941M except as follows (see 4.5):

a. The minimum time to first appearance of zinc alloy corrosion (white rust): 240 hours.
b. The minimum time to first appearance of base metal corrosion (red rust): 1000 hours.

3.6 Torque/Clamp Force. K-Factor shall be within the range indicated by Type on surrogate parts (see 1.3.1 and 4.6).

3.7 Adhesion. There shall be no blistering, peeling, flaking, or other separation of the zinc-nickel plating from the base metal (see 4.7).

3.8 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The final product shall contain no chemicals labelled “R1” or “R2” on the AD-DSL list. The valid AD-DSL will be the current English version on https://www.iaeg.com/chemicalrpt/addsl/. The content of the AD-DSL and its application does not relieve the plating applicator from obligation to comply with all existing relevant regional and national regulations (see 4.8).

3.9 Dry-to-touch. After application of the protective coating, the coated product shall be dry-to-touch (see 4.9).

3.10 Quality. As mentioned in SAE AMS2461, “plating, as received by the purchaser, shall be smooth, continuous, adherent to base metal, uniform in appearance and visually free from pin holes, porosity, blisters, nodules, pits, and other imperfections detrimental to usage of the plating. There shall be no evidence of electrical arcing or local overheating. Slight staining or discoloration is permissible” (reprinted with permission from SAE AMS2461; © SAE International. All rights reserved). The appearance may range from blue-bright to gray to silver. Color shall not be a reason for rejection (see 6.1).

4 VERIFICATIONS

4.1 Responsibility for compliance. All verifications in section 4 shall be the responsibility of the plating applicator (see 3.1 and Table I).

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TABLE I. Quality conformance verifications.

	Requirement	Verification	Sampling (see 4.1.4)	Sample Size (4.1.4)	Min. Test Frequency
Thickness	3.2	4.2	Production Part[1]	see 4.1.4	Each Lot
Plating process – HE Relief	3.2	4.2	Production Part[1,3]	2	Monthly[4]
Plating Composition	3.3	4.3	Production part, or Steel Test Panel, 0.6 mm (0.025 inch) min. thick	see 4.1.4	Quarterly[4]
Corrosion Resistance	3.5	4.5	Production Part[1,2]	2	Monthly[4]
Torque/Clamp Force (K-Factor)	3.6	4.6	Surrogate Part (4.6)	10	Daily[4]
Adhesion	3.7	4.7	Steel Test Panel, 0.6 mm (0.025 inch) min. thick	4	Daily[4]
Toxic chemicals	3.8	4.8	N/A		Upon Request
Dry-to-touch	3.9	4.9	Production Part		Upon Request
Quality	3.10	4.10	Production Part[1]	see 4.1.4	Each Lot

Note 1: Separate test specimens may only be used if: the plated parts are of such configuration or size as to be not readily adaptable to the specified test, when nondestructive testing is not practical, or it is not economically acceptable to perform destructive tests on actual parts.
Note 2: Production parts shall be used for corrosion testing during initial part submission. The use of steel parts or steel panels is acceptable for process control testing. Monthly process checks may omit the test for red corrosion provided an annual test for red corrosion is conducted.
Note 3: If required per ASTM F1941/F1941M hardness threshold requirements.
Note 4: Periodic testing may be suspended in any test period when parts are not processed but shall be performed before or at the time such processing is resumed.

4.1.1 Control records. At minimum, the plating applicator shall maintain records of verification per Table I.

4.1.2 Retaining records. The plating applicator shall maintain its records for a minimum of 3 years after final payment has been made for a procurement contract.

4.1.3 Lot. As defined in SAE AMS2461, “a lot is a group of parts, all of the same part number, processed through the same chemical solutions in the same tanks under the same conditions, which have completed the chemical processing within a period of 24 hours of each other and are presented to inspection at the same time” (reprinted with permission from SAE AMS2461; © SAE International. All rights reserved).

4.1.4 Lot sampling inspection. Sampling size shall be per Table II (reprinted with permission from SAE AMS2461; ©SAE International. All rights reserved). Exceptions to the lot sampling inspection are as indicated in Table I.

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TABLE II. Sampling for acceptance tests.

Number of Parts In Lot	Quality (see 4.10)	Thickness (see 4.2)
1 to 6	All or 3 (1)	3
7 to 15	7	4
16 to 40	10	4
41 to 110	15	5
111 to 300	25	6
301 to 500	35	7
501 to 700	50	8
701 to 1200	75	10
Over 1200	125	15

Notes:

(1) Whichever is less.

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4.2 Plating process. The plating applicator shall document that the alkaline electrodeposited coating meets the requirements of ASTM F1941/F1941M Fe/Zn-Ni 8BNS or 8BNSL. The coating thickness and hydrogen embrittlement relief, if applicable, shall also be documented (see 3.2 and Table I).

4.3 Plating composition. To determine alloy content, utilize X-Ray Fluorescence (XRF), spot-mode Energy Dispersive X-Ray (EDS), Atomic Absorption (AA), or other methods capable of measuring the stated composition weight requirement (see 3.3).

4.4 Sealant or topcoat. The presence of sealant or topcoat shall be verified by the torque/clamp force performance verification on surrogate parts (see 3.4 and 4.6)

4.5 Corrosion resistance. Corrosion resistance verification shall be conducted in accordance with ASTM F1941/F1941M (see 3.5). Low carbon steel panels shall be used for corrosion testing.

4.6 Torque/Clamp Force. Ten surrogate test parts shall be processed along with production parts through the entire plating and post-treatment process. These surrogate parts shall be processed and tested once per plating line, daily. After processing, the K-Factor shall be determined for the ten surrogate parts per ISO 16047, with the following clarifications (see 3.6 and 6.4):

a. Surrogate test bolts shall be property class 10.9 and meet the requirements of ISO 4014. The bolts shall be M10-1.5 X 60 or other length which allows for at least two full threads within the clamped length and at least two threads protruding through the test nut. Do not wash or pre-condition the surrogate test bolts after the coating application process and before torque/clamp force testing.
b. Test washers or test bearing plate(s) shall be according to ISO 16047 Type HH, plain (uncoated) surface, cleaned and degreased within 2 hours of the test. Surface roughness shall be less than Ra 0.8.
c. Test nut shall be an M10-1.5 according to ISO 4032 class 10, plain (uncoated) surface, cleaned and degreased within 2 hours of the test.
d. Speed of installation shall be 30 ± 5 RPM.

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4.7 Adhesion. Test and analyze using the heat/quench, bend, or burnishing test per ASTM B571. The choice of test method shall be one suitable for zinc or zinc alloy plating per ASTM B571 (see 3.7).

4.8 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. Upon request, the plating applicator (and their supplier(s), if any) shall certify the final product meets the requirement (see 3.8).

4.9 Dry-to-touch. After holding product with hand pressure for 5 to 10 seconds in a paper coffee filter or equivalent, there shall be no visible staining, as perceived with the unaided eye (see 3.9).

4.10 Quality. Using visual inspection, verify compliance with 3.10.

5 PACKAGING

This section is not applicable to this specification.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electrodeposited plating covered by this specification is intended for use as a corrosion protective coating. Color is NOT a requirement of this specification. However, yellow plating appearance is discouraged as this color implies the presence of hexavalent chromium in industry. Sealants or top coats may affect the underlying plating and conversion coating appearance. Typical color will range from gray to silver. Processes used for cleaning and deposition on parts heat-treated to core hardness greater than 39 HRC should be such that hydrogen embrittlement is minimized. Guidance is provided in ASTM F1941/F1941M.

6.1.1 Similarity to other specifications. This specification for zinc-nickel alloy finish is very similar to Type II, SAE AMS2461, “Plating, Zinc-Nickel Alloy.” Applicators that currently plate in accordance with AMS2461 may also meet the additional requirements of MIL-PRF-32647 by use of an appropriate sealant or topcoat to meet the corrosion and torque/clamp force requirements of this specification.

6.1.2 Fastener products. Specification ASME B18.24, Part Identifying Number (PIN) Code System Standard for B18 Products, includes this specification as a plating finish for steel fasteners. Plating PIN code QX corresponds to MIL-PRF-32647 Type K16, and QW corresponds to MIL-PRF-32647 Type K20.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Plating type (see 1.3.1)
   Example: “Zinc-nickel Electroplate, MIL-PRF-32647, Type K20: 0.18 – 0.22 K-Factor”
   “Zinc-nickel Electroplate, MIL-PRF-32647, Type K16: 0.14 – 0.18 K-Factor”

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6.3 General usage.

6.3.1 Temperatures. This plating should not be used on parts for Space applications or on parts which in service reach a temperature of 450°F (232°C) or higher or come in contact with other parts which reach those temperatures.

6.3.2 Paint adhesion. Users wishing to post-apply a coating to any part of the assembled fastener are cautioned that surface preparation and/or an adhesion test may be necessary.

6.4 Torque/Clamp Force. The torque/clamp force requirements listed are intended to provide the plating applicator with friction requirements for the plating system under specific conditions. In general, K-factors cannot be used to calculate torque values for production joints because K-factors change with fastener diameter, head type, mating parts, and other factors. Torque/Clamp Force testing is needed to determine the torque for a required clamp load of a production joint. Type K20 was originally established to provide similar torque/clamp force behavior as parts with zinc plating and a conversion coating containing hexavalent chromium. Type K20 is the suggested direct replacement. Type K16 was originally established as a direct replacement of cadmium plated parts. It is recommended the actual joint production assembly behavior be compared when making finish changes on any clamp load applications that are critical.

6.4.1 K-factor and coefficient of total friction relationship. ISO 16047, Fasteners - Torque/Clamp Force Testing, defines both the K-factor and the coefficient of total friction and describes methods for determining each. Essentially, the K-factor directly relates torque and tension and is impacted by the geometry of the test parts. That is why the surrogate test parts are precisely defined in section 4.6. The coefficient of total friction (CoF) can be calculated based on the geometry of these test parts. Using the relationships defined in ISO 16047, the approximate correlation is as follows:

CoF for Type K16: 0.10-0.14
CoF for Type K20: 0.13-0.17

6.5 Approved applicator list. It is encouraged that the purchaser of fasteners maintains a list of approved applicators per this specification (MIL-PRF-32647).

6.6 Non-Government publications. The year of issue of the specified publications at the time of publication of this standard will apply. Any conflict between this standard and a subsequent revision of a referenced publication should be brought to the attention of the Preparing Activity.

6.7 Definitions.

6.7.1 K-Factor. A dimensionless torque coefficient as described in ISO 16047.

6.7.2 Torque/Clamp Force. Relationship between torque and clamp force which can be quantified by the K-factor (see ISO 16047).

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6.8 Subject term (key word) listing.

Adhesion
Alkaline
Electroplate
K-Factor
Plating
Torque
Tension

6.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy – SH
Air Force – 70
DLA – GS

Review Activities:
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Air Force – 19

Civil agency:
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